

2024 AMC 10B Problem 5 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 5

Problem:

In the following expression, Melanie changed some of the plus signs to minus signs:

$$1 + 3 + 5 + 7 + \cdots + 97 + 99$$

When the new expression was evaluated, it was negative. What is the least number of plus signs that Melanie could have changed to minus signs?

Answer Choices:

- A. 14
- B. 15
- C. 16
- D. 17
- E. 18

Solution:

To minimize the number of minus signs needed to make the expression negative, minus signs should be chosen for all of the largest numbers. Hence the first k numbers of the expression will stay positive and the last $50 - k$ will be made negative for the greatest value of k that gives a negative value.

Recall that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$; that is, the sum of the first n odd positive integers is equal to n^2 . The expression in the problem statement is the sum of the first 50 odd positive integers, so it equals $50^2 = 2500$. Hence k^2 must be strictly less than $\frac{2500}{2} = 1250$. Because $35^2 = 1225$ and $36^2 = 1296$, at least (B) 15 plus signs must be switched to minus signs for the expression to evaluate to a negative value. Indeed,

$$1 + 3 + 5 + \cdots + 69 - 71 - 73 - 75 - \cdots - 99 = 1225 - (2500 - 1225) = -50$$

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